CrossTwin — System Overview

Django + PostGIS digital twin for Dutch urban data (EPSG:28992), with a separate TiTiler process for raster tiles

XYZ raster tiles (direct)

BROWSER

Map UI
Mapbox GL JS
Layers.js · mainMap.js
LstTool.js · SvfTool.js

Indicator panels
HTMX swaps (hx-get)
water · heat · housing · builtup
what-if sliders + switches

Map editing
Editing.js +
Mapbox GL Draw
(staff only)

Population dock
forecast curve +
what-if controls

Django admin
/admin/ (staff login)

JSON / HTML

HTMX

DJANGO · DIGITALTWIN :8000

URL router (DigitalTwin/urls.py)
/api/ · /watersupply/ · /housing/ · /urban_heat/ · /physicalEnv/ · /weather/ · /importer/

mainMap
layer catalog /api/layers/
GeoJSON + bounds endpoints
admin-unit & population panels
map editing (core/editing.py)

Domain apps
watersupply · housing · urban_heat
administrative · physicalEnv
builtup · nature · weather · Energy
on demand: LST (lst/) · SVF

core
MODEL / VECTOR / RASTER /
WMS registries (utils.py)
rasterOperations → COG
signals · cache versioning

importer
file upload (GeoJSON/SHP)
external catalog (70 sets)
SpatialParentIndex
BulkWriter · deferred_cascades

weather WMS proxy
/weather/wms/<name>/tile/
keep-alive session
24 h cache for TIME= frames

Caches (LocMem)
geojson — 2 min, ≤ 8 MB
wms_tiles — 24 h / 5 min
default

Signals keep derived data consistent: population & urban-area cascade, watersupply indicator chain, raster → COG export, layer cache bumps.

/cog/info

TiTiler :8001
FastAPI · tiler.py
/cog/tiles/{z}/{x}/{y}
/cog/info

read COG

COG store
cogs/<app_label>/*.tif
EPSG:4326, rio-cogeo
written by post_save

fetch datasets

WMS GetMap

EXTERNAL

- PDOK
- CBS
- BIS Nederland
- OpenStreetMap
- Sentinel-2
- Earth Engine
- KNMI
- RIVM
- Klimaateffectatlas
- WMS servers

ORM + raw SQL

PostgreSQL 16 + PostGIS
84 models · GeometryField & RasterField (srid 28992) · GiST indexes · pg_stat_statements

Browser · Django app · Raster / tiles · Cache · Storage · External source